TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	64.4 ± 2.62
Correlation Stability	87.2 ± 0.75
Distribution Consistency	79.3 ± 7.52
Entropy Preservation	29.1 ± 2.55
Mean Z Score Drift	0.4 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	5.6 ± 1.62
Data Density	100.0 ± 0.00
Coverage Score	87.9 ± 1.92
Feature Accuracy	62.9 ± 0.94
Novelty Score	0.7 ± 0.02

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	56.886	+2.447	4.5%	OK
sex	0.680	0.640	-0.040	5.8%	OK
cp	3.158	3.501	+0.343	10.9%	Warn
trestbps	131.690	136.573	+4.883	3.7%	OK
chol	246.693	251.089	+4.396	1.8%	OK
fbs	0.149	0.037	-0.111	74.9%	Drift
restecg	0.990	1.037	+0.047	4.8%	OK
thalach	149.607	135.774	-13.833	9.2%	OK
exang	0.327	0.303	-0.024	7.3%	OK
oldpeak	1.040	1.762	+0.723	69.5%	Drift
slope	1.601	1.690	+0.089	5.6%	OK
ca	0.672	1.339	+0.667	99.2%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	64.4 ± 2.62
Correlation Stability (*2)	87.2 ± 0.75
Distribution Consistency (*3)	79.3 ± 7.52
Entropy Preservation (*4)	29.1 ± 2.55
Mean Z Score Drift (*5)	0.4 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	5.6 ± 1.62

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	87.9 ± 1.92
Feature Accuracy (*10)	62.9 ± 0.94
Novelty Score (*11)	0.7 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.16557377049180327	0.5557377049180328	-0.05245901639344264	
F1	0.6070236204175147	0.1104621946906585	0.5619870332268747	-0.04503658719063984	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5770491803278689	0.15737704918032785	0.5459016393442624	-0.03114754098360658	*
F1	0.5344485042532131	0.10439358158687953	0.5397081118369083	0.005259607583695136	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5360655737704918	0.15081967213114753	0.5213114754098365	-0.014754098360655754	
F1	0.5219249758934431	0.10103233552886673	0.5199244286019669	-0.002000547291476090 3	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5704918032786885	0.14590163934426233	0.4786885245901638 7	-0.0918032786885246	*
F1	0.5303766679506219	0.09243364009669637	0.4976555862697702	-0.03272108168085167	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5983606557377049	0.15245901639344264	0.5147540983606558	-0.0836065573770492	*
F1	0.5648347697855012	0.09948564664283634	0.5343325685846874	-0.03050220120081413	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.05245901639344262	0.4229508196721311	-0.12459016393442623	*
F1	0.3899270600110827	0.03318573615109445	0.4530531760379284	0.06312611602684572	*

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5426229508196723	0.14098360655737702	0.5672131147540983	0.02459016393442622	
F1	0.5198013329305267	0.09418556832947925	0.5626214364332811	0.04282010350275442	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.12786885245901644	0.5065573770491804	-0.05081967213114753	*
F1	0.568660979990255	0.07094961106654392	0.5340038124531278	-0.03465716753712735	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5295081967213116	0.15245901639344261	0.4852459016393443	-0.04426229508196723	
F1	0.5368793679425432	0.10277380150139655	0.5053073241376916	-0.03157204380485150	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.1	0.3147540983606557	-0.23934426229508202	*
F1	0.4442202071436568 4	0.07171053873757706	0.3651927458319015 3	-0.07902746131175528	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	56.0	1.0	4.0	130.0	251.0	0.0	0.0	143.0	1.0	1.131041217253625
2	48.0	1.0	2.0	119.0	184.0	0.0	1.0	159.0	0.0	0.3235241944603261
3	59.0	1.0	1.0	169.0	287.0	0.0	2.0	158.0	0.0	0.19871449847882072
4	57.0	1.0	4.0	151.0	274.0	0.0	0.0	94.0	1.0	1.0414362489042723
5	47.0	1.0	4.0	125.0	291.0	0.0	2.0	164.0	0.0	0.0
6	64.0	1.0	3.0	146.0	209.0	0.0	2.0	144.0	0.0	1.2832705136118916
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	49.0	1.0	2.0	110.0	229.0	0.0	0.0	164.0	0.0	0.5680753729212469
2	64.0	1.0	3.0	126.0	311.0	0.0	0.0	133.0	0.0	1.6284876091899327
3	57.0	1.0	2.0	129.0	223.0	0.0	1.0	152.0	0.0	0.4808090060595247
4	59.0	1.0	3.0	136.0	212.0	0.0	2.0	141.0	0.0	1.4970154442831078
5	60.0	0.0	3.0	147.0	305.0	0.0	2.0	147.0	0.0	0.9291319303378895
6	61.0	0.0	4.0	143.0	308.0	0.0	2.0	142.0	0.0	1.137558428467799
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	49.0	0.0	4.0	115.0	181.0	0.0	1.0	149.0	0.0	0.0
2	55.0	0.0	4.0	136.0	201.0	0.0	1.0	150.0	1.0	2.4416913786599013
3	64.0	0.0	2.0	124.0	275.0	0.0	1.0	171.0	0.0	1.5861480026961463
4	57.0	0.0	4.0	151.0	273.0	0.0	0.0	88.0	1.0	1.1716451862544166
5	53.0	1.0	4.0	126.0	254.0	0.0	0.0	160.0	0.0	0.04812517135275329
6	50.0	1.0	3.0	111.0	229.0	0.0	0.0	163.0	0.0	0.4550251434432545
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	55.0	1.0	3.0	130.0	262.0	0.0	0.0	164.0	0.0	0.3646141418719901 5
2	61.0	1.0	4.0	121.0	255.0	0.0	2.0	144.0	0.0	2.2939010326747886
3	43.0	1.0	3.0	112.0	165.0	0.0	2.0	148.0	0.0	0.2260370012247391 3
4	57.0	1.0	4.0	127.0	250.0	0.0	1.0	144.0	1.0	1.2961045216683544
5	64.0	1.0	1.0	135.0	282.0	0.0	2.0	172.0	0.0	1.4468872671413653
6	65.0	1.0	2.0	148.0	210.0	0.0	1.0	152.0	0.0	0.8
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	54.0	1.0	1.0	188.0	283.0	0.0	2.0	188.0	0.0	0.0349272694506285
2	57.0	0.0	2.0	138.0	234.0	0.0	2.0	170.0	0.0	0.0
3	62.0	0.0	4.0	124.0	199.0	0.0	0.0	135.0	1.0	0.3084009836178004
4	49.0	1.0	2.0	122.0	283.0	0.0	2.0	158.0	0.0	1.0996584207270061
5	58.0	0.0	4.0	131.0	199.0	0.0	0.0	144.0	1.0	1.4628538358519645
6	57.0	1.0	3.0	128.0	228.0	0.0	2.0	150.0	0.0	0.4124692307900879
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	59.0	0.0	1.0	165.0	291.0	0.0	2.0	156.0	0.0	0.3456860046592206
2	64.0	1.0	4.0	110.0	247.0	0.0	1.0	158.0	0.0	0.5791333143837275
3	58.0	1.0	2.0	157.0	273.0	0.0	1.0	111.0	0.0	0.4442388608336034
4	69.0	1.0	3.0	129.0	321.0	0.0	1.0	109.0	0.0	2.3777084282490186
5	55.0	0.0	4.0	135.0	312.0	0.0	0.0	171.0	0.0	0.9267925442405431
6	56.0	1.0	4.0	125.0	249.0	0.0	1.0	144.0	1.0	1.2220879300811955
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	54.0	0.0	2.0	133.0	234.0	0.0	1.0	169.0	0.0	0.2233071704967825
2	59.0	0.0	3.0	147.0	241.0	0.0	0.0	156.0	0.0	1.2148919415095703
3	57.0	0.0	4.0	148.0	263.0	0.0	0.0	99.0	1.0	0.8801301994876438
4	64.0	1.0	3.0	125.0	309.0	0.0	0.0	131.0	0.0	1.7505735179599164
5	57.0	1.0	2.0	120.0	281.0	0.0	1.0	160.0	0.0	1.6365875300993302
6	41.0	1.0	4.0	110.0	178.0	0.0	2.0	162.0	0.0	0.0
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	62.0	0.0	4.0	124.0	200.0	0.0	0.0	135.0	1.0	0.3542090448508086
2	52.0	0.0	3.0	141.0	234.0	0.0	0.0	161.0	0.0	0.7428953618877427
3	44.0	1.0	4.0	114.0	181.0	0.0	2.0	151.0	0.0	0.4996511755331646
4	44.0	1.0	3.0	138.0	241.0	0.0	1.0	144.0	0.0	1.2904324388799275
5	56.0	1.0	3.0	124.0	254.0	0.0	1.0	142.0	0.0	0.7687953794613424
6	51.0	1.0	3.0	111.0	230.0	0.0	0.0	159.0	0.0	0.0
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	64.0	1.0	1.0	136.0	282.0	0.0	2.0	172.0	0.0	1.4314307460815439
2	61.0	0.0	4.0	119.0	261.0	0.0	1.0	150.0	1.0	2.4495471763607832
3	49.0	1.0	2.0	115.0	237.0	0.0	0.0	165.0	0.0	0.6831551832771283
4	58.0	1.0	4.0	127.0	251.0	0.0	0.0	144.0	1.0	1.3169924589145063
5	58.0	1.0	1.0	133.0	208.0	0.0	0.0	160.0	0.0	0.7342325237634215
6	51.0	0.0	3.0	119.0	158.0	0.0	1.0	127.0	0.0	0.6428727765739315
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	47.0	1.0	4.0	111.0	274.0	0.0	2.0	117.0	1.0	1.1010682999639254
2	58.0	1.0	2.0	121.0	288.0	0.0	2.0	162.0	0.0	1.3393213252873588
3	60.0	0.0	4.0	129.0	327.0	0.0	2.0	169.0	0.0	0.0
4	48.0	0.0	2.0	112.0	229.0	0.0	0.0	168.0	0.0	0.8911779892563932
5	47.0	1.0	3.0	119.0	176.0	0.0	2.0	157.0	0.0	0.9609996185719156
6	49.0	1.0	4.0	114.0	272.0	0.0	2.0	115.0	1.0	1.3719683817038886
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6